

HASSAN SHAHID

Data Analyst | Python | SQL | Power BI | Machine Learning
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PROFESSIONAL SUMMARY

Business Data Analytics student at COMSATS University Islamabad with hands-on experience turning raw data into business decisions using Python, SQL, and machine learning. Built a fraud-detection model catching 86% of fraud cases at 95% precision across 284,807 transactions, and delivered a validated ARIMA forecasting pipeline for PSX equity prices with full statistical diagnostics. Experienced in a regulated compliance environment applying AML/KYC frameworks to real financial data. Certified in Python (Cisco Networking Academy) and Microsoft Office.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn), SQL (Oracle), HTML5, CSS

Analytics & ML: Exploratory Data Analysis (EDA), Logistic Regression, ARIMA, Time Series Forecasting, Feature Engineering, Outlier Detection, Data Cleaning, Statistical Testing (ADF, Ljung-Box)

Tools & Platforms: Power BI (DAX, Dashboards), Jupyter Notebook, Microsoft Excel (Pivot Tables, Charts), PowerPoint, Yahoo Finance API, Kaggle, Git

Soft Skills: Attention to Detail, Compliance & AML/KYC Awareness, Team Collaboration, Problem-Solving

PROJECTS

Credit Card Fraud Detection | *Python • Pandas • Scikit-learn • Matplotlib • Seaborn*

- Analysed 284,807 real-world credit card transactions (Kaggle); identified 492 fraud cases (0.17%) and addressed severe class imbalance using 10:1 undersampling to build a balanced training set.
- Applied IQR outlier capping across all 31 features and log transformation (np.log1p) on transaction amounts to normalize skewed distributions before model training.
- Built a fraud-detection model catching 86% of fraudulent transactions (118/137) at 95% precision, minimizing false-positive investigation load while trained on 284K real transactions with severe (0.17%) class imbalance.
- Produced 8 visualizations — confusion matrix heatmap, correlation heatmap, scatter plots (V1 vs V2), boxplots, and moving averages — to communicate data insights clearly.

ENGRO Stock Price Forecasting — ARIMA Model | *Python • Statsmodels • Pandas • Yahoo Finance API*

- Collected 3+ years of ENGRO daily closing prices from the Pakistan Stock Exchange (PSX) via the yFinance API; visualized trend, volatility, and seasonality patterns in the time series.
- Confirmed non-stationarity using the Augmented Dickey-Fuller (ADF) test; applied first-order differencing and re-tested to satisfy the integration (I) component of ARIMA.
- Identified optimal ARIMA(p,d,q) parameters by interpreting ACF and PACF plots; estimated model and validated using AIC/BIC selection criteria and Ljung-Box residual test (white-noise confirmed).
- Delivered a validated 10–30 day price forecasting pipeline with quantified confidence intervals, enabling data-driven horizon selection and clearly documenting model reliability limits for investment decision-making.

MediGuard — Medicine Expiry Tracker | *Python • Pandas*

- Automated expiry tracking across a pharmaceutical inventory dataset with Pandas, surfacing near-expiry stock before it became loss and cutting manual audit checks from a daily task to a single automated report.
- Implemented date comparison logic, conditional filtering, and tabular reporting to deliver actionable alerts from pharmaceutical datasets.

WORK EXPERIENCE

Compliance & Support Officer • RemitHome Oct 2025 | Part-time

- Reviewed and verified [X]+ daily remittance transactions under AML/KYC compliance frameworks with zero missed regulatory flags, ensuring accuracy and timely completion in a data-sensitive environment.
- Investigated and documented flagged transaction activity, feeding directly into audit trail integrity and demonstrating data quality control in a regulated financial environment.
- Coordinated across teams to resolve customer queries efficiently, demonstrating communication and attention to detail in a data-sensitive environment.

EDUCATION

Bachelor of Science in Business Data Analytics 2024 – Present
COMSATS University Islamabad

Intermediate in Computer Science (ICS) 2021 – 2023
Punjab Group of Colleges, Faisalabad

CERTIFICATIONS

- Python Essentials 1 — Cisco Networking Academy / COMSATS University Islamabad (Jun 2025) | Cert ID: a0fb21a0-c7eb-4c3f-bb76-04407d436b24
- Microsoft Office Specialist: Associate (Office 2019) — Microsoft